📊 RURO Post-Estimation Analysis

Generated: 2026-05-13 12:10:58

Total Parameters: 52 | **Groups:** joint

Estimation Results Source:

`U:\Desktop\Nizam_Hisham\MNL\outputs\estimates\fr\spec\stijn_occ\gamspy\estimation_spec_stijn_occ_M0\run_2026-05-13_11-27-40\estimation_results.json`

Estimation Configuration

Specification	stijn_occ_M0
Specification File	`U:\Desktop\Nizam_Hisham\MNL\scripts\enhanced\estimation_spec_stijn_occ_M0.yaml`
Model Family	regular
Proposal Correction	Enabled
Proposal Correction Form	-log(prior)
Opportunity Centering	Enabled

Single Males Single Females Males in Couples Females in Couples

⏱ Elapsed Time

Estimation	Post-Estimation	Total	Iterations
5m 0s	**20.8s**	**5m 21s**	**29**

📈 Model Fit Statistics

log_likelihood	-6499.8809
n_observations	4.2530e+05

n_groups	4253.0000
n_parameters	52.0000
AIC	1.3104e+04
BIC	1.3674e+04
ll_null	-1.9586e+04
ll_null_uniform	-1.9586e+04
ll_null_prior_corrected	-2.2322e+04
n_obs_long	4.2530e+05
rho_squared_uniform	0.6681
rho_squared_adj_uniform	0.6655
rho_squared_prior_corrected	0.7088
rho_squared_adj_prior_corrected	0.7065
rho_squared	0.7088
rho_squared_adj	0.7065
AIC_per_obs	0.0308
n_negative_muc_total	0.0000e+00
n_negative_mul_total	0.0000e+00
pct_negative_muc_total	0.0000e+00
pct_negative_mul_total	0.0000e+00
n_bounded_params	52
n_hit_lower_bound	0
n_hit_upper_bound	0

📝 Understanding Bounded Parameters

- **n_bounded_params (52)**: Parameters with constraints defined.
- **n_hit_lower_bound (0)**: Parameters at lower bound. ✓ Good!

- **n_hit_upper_bound (0)**: Parameters at upper bound. ✓ Good!

🔍 Identification Diagnostics

Hessian-based diagnostics for parameter identification quality.

Condition Number: 6.76e+10

✗ Severe Identification Problem

The Hessian is severely ill-conditioned (κ ≥ 10,000). This indicates serious identification issues with unreliable parameter estimates.

Eigenvalue Analysis

Minimum Eigenvalue

-2.60e+01

Maximum Eigenvalue

1.42e+10

Negative Eigenvalues

2

✗ NOT a local maximum!

Eigenvalue Summary

p5 / median / p95: 9.74e-01 / 1.55e+02 / 7.02e+05

Smallest 5: -2.60e+01, -2.10e-01, 7.92e-01, 1.12e+00, 1.25e+00

Largest 5: 5.01e+05, 5.43e+05, 8.97e+05, 1.03e+06, 1.42e+10

High Correlations (|rho| >= 0.90)

Parameter A	Parameter B	Correlation
`beta_c_sf`	`theta_c_sf`	-1.035
`beta_l0_sf`	`beta_l_educH_sf`	+0.996
`beta_c_sm`	`theta_c_sm`	+0.988
`beta_l0_f`	`beta_l_educH_f`	+0.979

Parameter A	Parameter B	Correlation
`beta_l0_sm`	`beta_l_educH_sm`	+0.971
`beta_w_pexp`	`beta_w_pexp2`	-0.961
`beta_E`	`beta_E_gsur`	-0.947
`beta_c`	`theta_c`	+0.935
`beta_c_sm`	`beta_c`	-0.918
`theta_c_sm`	`beta_c`	-0.916

ℹ️ **Technical Notes**

Condition Number (κ): Measures sensitivity of parameter estimates to small changes in data. $\kappa = \lambda_{max} / \lambda_{min}$, where λ are eigenvalues of -H (Hessian).

Eigenvalues: All eigenvalues of -H should be positive for a local maximum. Near-zero eigenvalues indicate flat directions (weak identification).

Sign convention: Eigenvalues shown come from the Hessian used to compute SEs. When SEs are computed numerically, this is the Hessian of the negative log-likelihood (H), so negative values indicate H is not positive semidefinite (ill-conditioning or non-optimum).

Standard Errors: Computed as $SE = \sqrt{\mathrm{diag}((-H)^{-1})}$, where H is the Hessian at the optimum. Large SE indicates parameter is not precisely estimated.

🧪 Identification Diagnostics (Estimation Stage)

Source:

`U:\Desktop\Nizam_Hisham\MNL\outputs\estimates\fr\spec\stijn_occ\gamspy\estimation_spec_stijn_occ_M0\run_2026-05-13_11-27-40\identification_diagnostics.txt`

```
Identification Diagnostics
================================================================================
specification: stijn_occ_M0
n_parameters: 52
bounded_hits_total: 0
bounded_hits_lower: 0
bounded_hits_upper: 0
```

```
hessian_condition_number: 67638896031.10768
hessian_negative_eigenvalues: 2
hessian_near_zero_eigenvalues(|eig|<=1e-8): 0
negative_variances_from_varcov: 2

Parameter Stability
--------------------------------------------------------------------------------
reference: initial_vector
matched_params: 52/52
delta_l2: 1.080135e+01
delta_l2_relative: 2.524920e+00
delta_max_abs: 4.758525e+00
delta_mean_abs: 8.790383e-01

Bound Hits
--------------------------------------------------------------------------------
none
```

📐 Model Specification

Utility and opportunity functions for each demographic group, shown in both symbolic and numerical form.

Single Males

📐 Utility Function (Symbolic)

$$U = \beta_C \cdot BC(C, \theta_C) + \beta_l(X) \cdot BC(L, \theta_l)$$

where $BC(x, \theta) = (x^\theta - 1) / \theta$ (Box-Cox transformation)

$\beta_l(X) = \beta_{l0} + \beta_{l,age} \cdot age + \beta_{l,age2} \cdot age2 + \beta_{l,educH} \cdot educH$

🔢 Utility Function (Numerical)

$$U = 0.7175 \cdot BC(C, -0.8560) + (4.4666 + 0.0008 \cdot age + 0.0018 \cdot age2 - 1.3035 \cdot educH) \cdot BC(L, -0.7038)$$

Single Females

📐 **Utility Function (Symbolic)**

$$U = \beta_c \cdot BC(C, \theta_c) + \beta_l(X) \cdot BC(L, \theta_l)$$

where $BC(x, \theta) = (x^{\theta} - 1) / \theta$ (Box-Cox transformation)
$\beta_l(X) = \beta_{l0} + \beta_{l,age} \cdot age + \beta_{l,age2} \cdot age2 + \beta_{l,nkids} \cdot nkids + \beta_{l,educH} \cdot educH$

🔢 **Utility Function (Numerical)**

$$U = 0.4630 \cdot BC(C, -1.0884) + (5.7585 - 0.0165 \cdot age + 0.0033 \cdot age2 - 0.0496 \cdot nkids - 2.2710 \cdot educH) \cdot BC(L, -0.7238)$$

Males in Couples

📐 **Utility Function (Symbolic)**

$$U = \beta_c \cdot BC(C, \theta_c) + \beta_l(X) \cdot BC(L, \theta_l)$$

where $BC(x, \theta) = (x^{\theta} - 1) / \theta$ (Box-Cox transformation)
$\beta_l(X) = \beta_{l0} + \beta_{l,age} \cdot age + \beta_{l,age2} \cdot age2 + \beta_{l,educH} \cdot educH$

🔢 **Utility Function (Numerical)**

$$U = 5.2620 \cdot BC(C, 0.2153) + (2.7023 - 0.0074 \cdot age + 0.0013 \cdot age2 - 0.9259 \cdot educH) \cdot BC(L, -0.7330)$$

Females in Couples

📐 Utility Function (Symbolic)

$$U = \beta_c \cdot BC(C, \theta_c) + \beta_l(X) \cdot BC(L, \theta_l)$$

where BC(x, θ) = (x^θ - 1) / θ (Box-Cox transformation)
$\beta_l(X) = \beta_{l0} + \beta_{l,age} \cdot age + \beta_{l,age2} \cdot age2 + \beta_{l,nkids} \cdot nkids + \beta_{l,educH} \cdot educH$

🔢 Utility Function (Numerical)

$$U = 5.2620 \cdot BC(C, 0.2153) + (5.6972 - 0.0578 \cdot age + 0.0016 \cdot age2 + 0.2149 \cdot nkids - 1.4890 \cdot educH) \cdot BC(L, -0.6957)$$

📐 Model Index

$$V_{ij} = U_{ij} + O^E_{ij} + O^H_{ij} + O^W_{ij} + O^{Occ}_{ij} + - \log prior_{ij}$$

$$P_{ij} = \exp(V_{ij}) / \Sigma_{k\in C_i} \exp(V_{ik})$$

U is the utility of consumption and leisure; the O˙ terms are opportunity-density contributions (employment/hours, wage, occupation); `log prior` rescales each alternative to the data-generating measure. The terms displayed reflect the opportunity blocks declared in the active YAML specification.

⏰ Employment and Hours Opportunity Parameters

Working-gated employment and hours intercepts. Combines `hours_opportunity.shifters` (base employment intercept and PT/FT focal-point indicators) with `market_opportunity.shifters` (working-gated residual market access terms like gsur, education).

Symbolic form

$$O^E + O^H =$$

```
               beta_E · working
  + beta_h_pt1 · working_pt1
  + beta_h_pt2 · working_pt2
  + beta_h_ft · working_ft
  + beta_E_gsur · gsur · working
  + beta_E_educH · educH · working
```

Numerical form (by group)

Joint (All Groups): -2.6148 · working + -0.5203 · working_pt1 + 0.3721 · working_pt2 + 1.4632 · working_ft + -0.7719 · gsur · working + 0.2575 · educH · working

💰 Wage Opportunity / Mincer Parameters

Log-normal wage opportunity: `log(wage) = μw(X) + ε, ε ~ N(0, σ²)`. The mean μ_w is built from `wage_opportunity.mean_shifters`.

Symbolic form

$$\mu_w =$$

```
          beta_w0
+ beta_w_educL  · educL
+ beta_w_educH  · educH
+ beta_w_pexp   · pexp_years
+ beta_w_pexp2  · pexp_years2
```

$$\log(\text{wage}) = \mu_w + \varepsilon, \quad \varepsilon \sim N(0, \text{sigma}^2)$$

Numerical form (by group)

Joint (All Groups): $\mu_w = 2.0411 + -0.0476 \cdot \text{educL} + 0.3055 \cdot \text{educH} + 0.0174 \cdot \text{pexp\_years} + -0.0002 \cdot \text{pexp\_years2}$

`sigma = 0.4232`

🧭 Occupation Opportunity Parameters

Working-gated occupation shifters. Each `applies_to` group has its own coefficient set on the non-reference categories of `loc4`. Reference category: `loc4 = 1`.

Single Males (`applies_to = sm`)

$$O^{Occ}_{sm} =$$

```
beta_occ_2_sm  · loc4_2  · working
```

$$+ \text{beta\_occ\_3\_sm} \cdot \text{loc4\_3} \cdot \text{working} + \text{beta\_occ\_4\_sm} \cdot \text{loc4\_4} \cdot \text{working}$$

$$-1.5129 \cdot \text{loc4\_2} \cdot \text{working} + -2.1726 \cdot \text{loc4\_3} \cdot \text{working} + 0.0175 \cdot \text{loc4\_4} \cdot \text{working}$$

Single Females (`applies_to = sf`)

$$O^{Occ}_{sf} =$$

$$\text{beta\_occ\_2\_sf} \cdot \text{loc4\_2} \cdot \text{working} + \text{beta\_occ\_3\_sf} \cdot \text{loc4\_3} \cdot \text{working} + \text{beta\_occ\_4\_sf} \cdot \text{loc4\_4} \cdot \text{working}$$

$$-0.0147 \cdot \text{loc4\_2} \cdot \text{working} + -0.5745 \cdot \text{loc4\_3} \cdot \text{working} + 0.7934 \cdot \text{loc4\_4} \cdot \text{working}$$

couples_male (`applies_to = cm`)

$$O^{Occ}_{cm} =$$

$$\text{beta\_occ\_2\_cm} \cdot \text{loc4\_2} \cdot \text{working} + \text{beta\_occ\_3\_cm} \cdot \text{loc4\_3} \cdot \text{working} + \text{beta\_occ\_4\_cm} \cdot \text{loc4\_4} \cdot \text{working}$$

$$-1.4707 \cdot \text{loc4\_2} \cdot \text{working} + -2.2061 \cdot \text{loc4\_3} \cdot \text{working} + 0.4862 \cdot \text{loc4\_4} \cdot \text{working}$$

couples_female (`applies_to = cf`)

$$O^{Occ}_{cf} =$$

```
          beta_occ_2_cf · loc4_2 · working
+ beta_occ_3_cf · loc4_3 · working
+ beta_occ_4_cf · loc4_4 · working
```

```
0.2050 · loc4_2 · working + -0.1904 · loc4_3 · working + 1.1460 · loc4_4 · working
```

📈 Curvature-Based Heuristics (Structural Elasticity Approximations)

⚠️ **Interpretation Note**

These are **not** true labor supply elasticities. They are heuristic approximations derived from the curvature parameters (θ) of the Box-Cox utility function. The Hicksian approximation is (1 - θ_l), which measures the curvature of preferences. For rigorous elasticity estimates, use simulation-based methods that account for the full discrete choice structure and budget constraints.

📊 Labor Supply Elasticities

Group	Hicksian (compensated)	Marshallian (uncompensated)	Participation (extensive)	Intensive (conditional)	θ_l	θ_c	β_l (at median X)	β_c
Single Males	1.704	1.604	0.511	1.193	-0.704	-0.856	4.467	0.718

Group	Hicksian (compensated)	Marshallian (uncompensated)	Participation (extensive)	Intensive (conditional)	θ_l	θ_c	β_l (at median X)	β_c
Single Females	1.724	1.624	0.517	1.207	-0.724	-1.088	5.759	0.463
Males in Couples	1.733	1.633	0.520	1.213	-0.733	0.215	2.702	5.262
Females in Couples	1.696	1.596	0.509	1.187	-0.696	0.215	5.697	5.262

🎯 Fit Diagnostics

Observed vs Predicted

Group	Obs. Participation	Pred. Participation	Obs. Mean Hours	Pred. Mean Hours
Single Males	94.0%	100.0%	36.3	34.4
Single Females	93.0%	100.0%	39.3	34.1
Males in Couples	97.2%	100.0%	41.6	58.6
Females in Couples	96.5%	100.0%	35.6	58.1

⚠️ Marginal Utility Diagnostics

Analysis at chosen alternatives. Negative values indicate potential specification issues.

Negative MUC/MUL Summary

MUC Behavior Analysis

For well-behaved utility: MUC > 0 (β_c > 0) and diminishing (θ_c < 1)

Group	N Individuals	% Neg MUC	% Neg MUL	MUC Mean	MUL Mean
Single Males	910	0.00%	0.00%	0.0000	7.0339e-03
Single Females	766	0.00%	0.00%	0.0000	9.6380e-03
Males in Couples	2577	0.00%	0.00%	0.0079	5.4326e-03
Females in Couples	2577	0.00%	0.00%	0.0079	9.0155e-03

Group	β_c	θ_c	MUC Positive?	MUC Diminishing?	Well-Behaved?	MUC at Median C	C where MUC=1	Notes
Single Males	0.7175	-0.8560	✓	✓	✓	0.7175	0.8362	
Single Females	0.4630	-1.0884	✓	✓	✓	0.4630	0.6916	
Males in Couples	5.2620	0.2153	✓	✓	✓	5.2620	8.2988	
Females in Couples	5.2620	0.2153	✓	✓	✓	5.2620	8.2988	

📊 Choice Probability Diagnostics

Probability Sum Sanity Check

Max |Σp - 1|	0.000000
Mean |Σp - 1|	0.000000
% HH off by >0.01	0.00%
% HH off by >0.001	0.00%

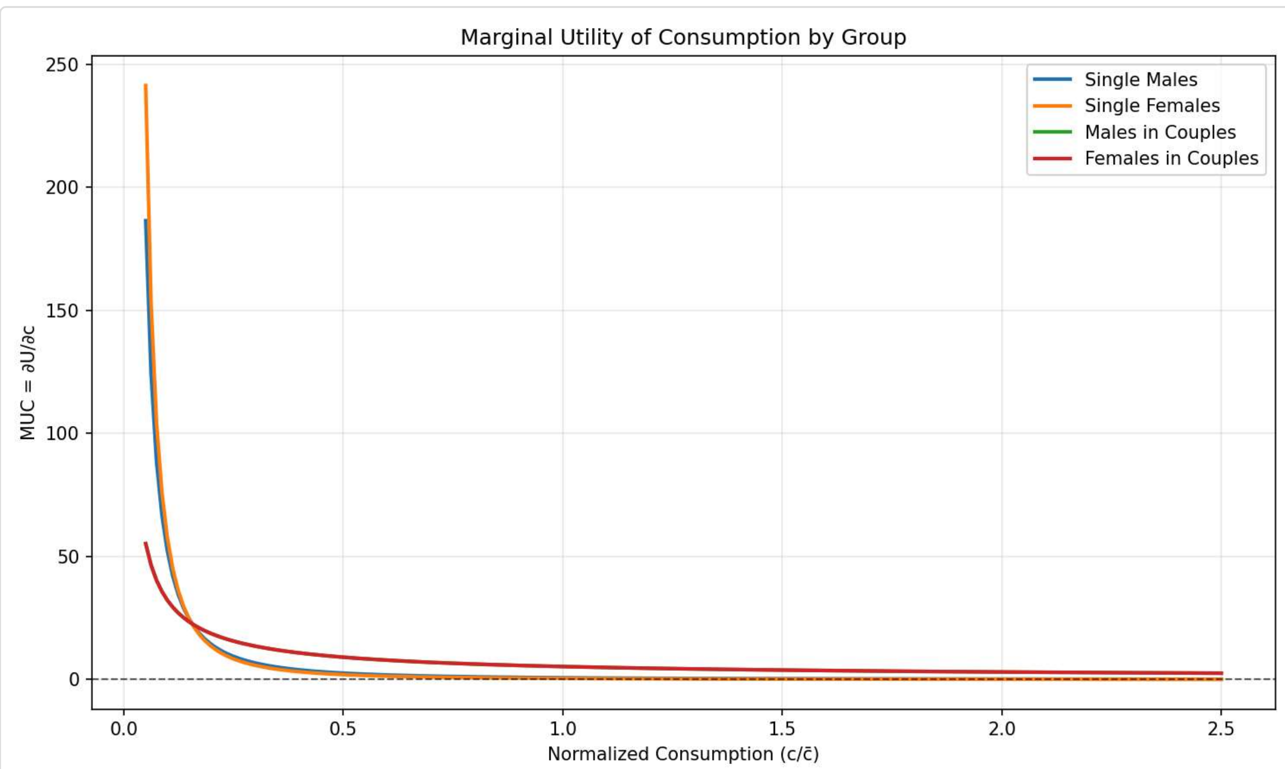
P(chosen) Distribution

Min	0.0000
10th percentile	0.0130
25th percentile	0.0707
Median	0.2547
Mean	0.3707
75th percentile	0.6810
90th percentile	0.9085
Max	0.9998

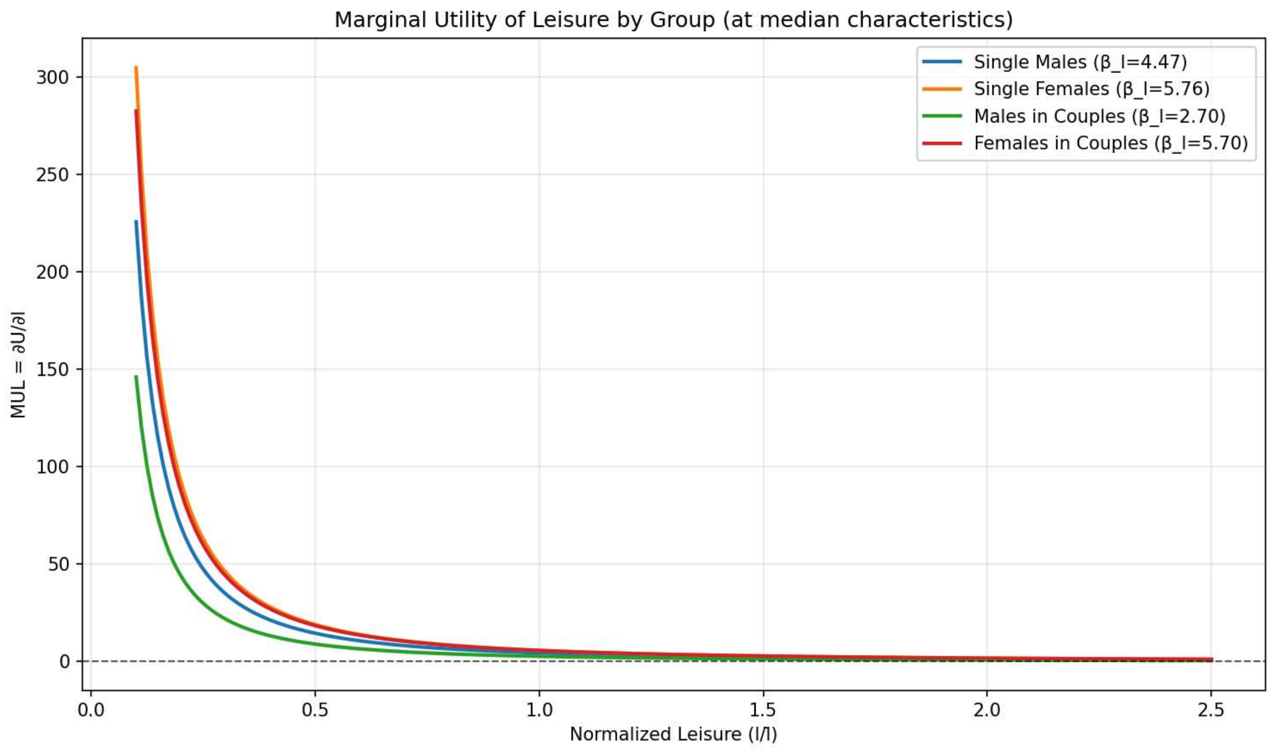
🗺️ Utility Contours & Plots

Marginal Utility Comparison

Marginal Utility of Consumption

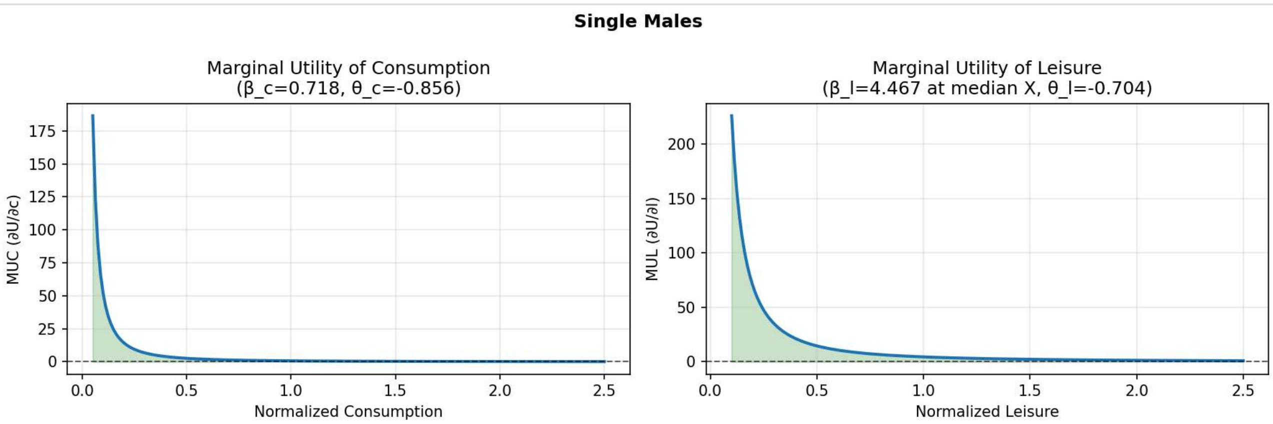


Marginal Utility of Leisure

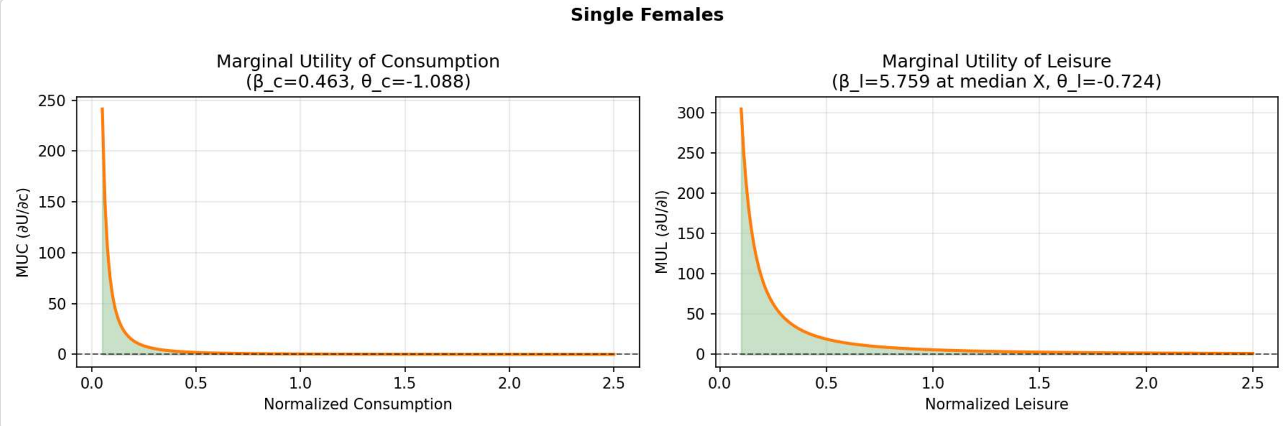


📊 Marginal Utility Distributions by Group

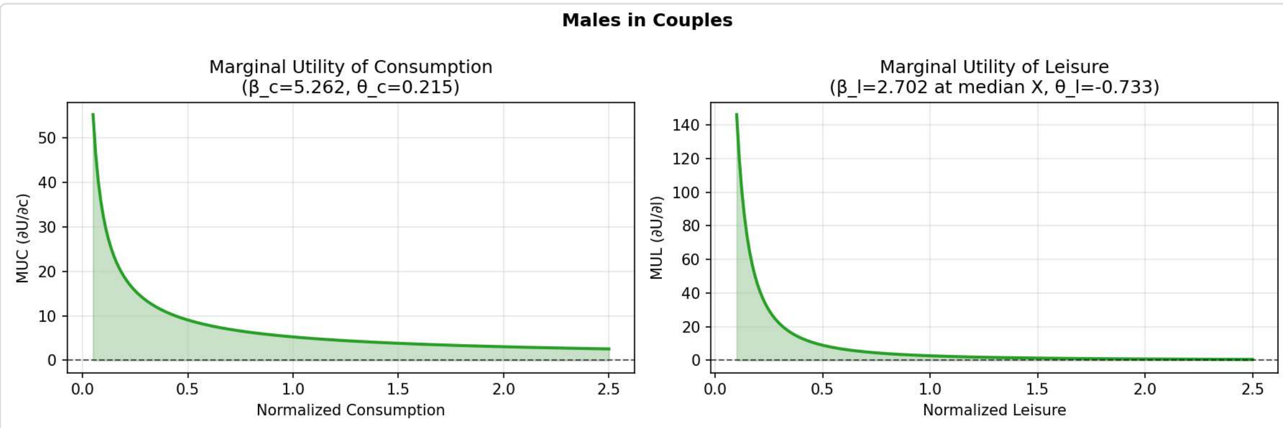
Single Males - MUC & MUL Distributions



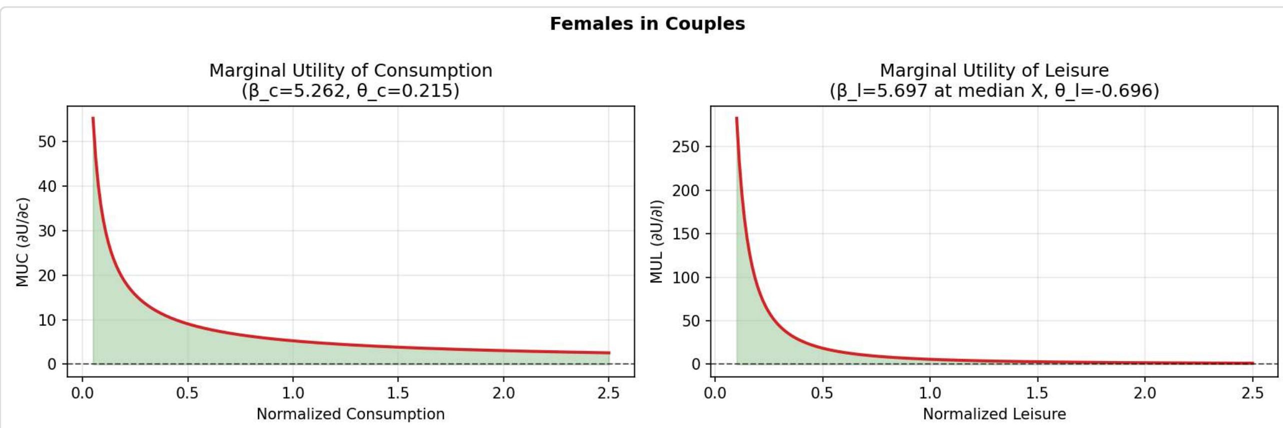
Single Females - MUC & MUL Distributions



Males in Couples - MUC & MUL Distributions

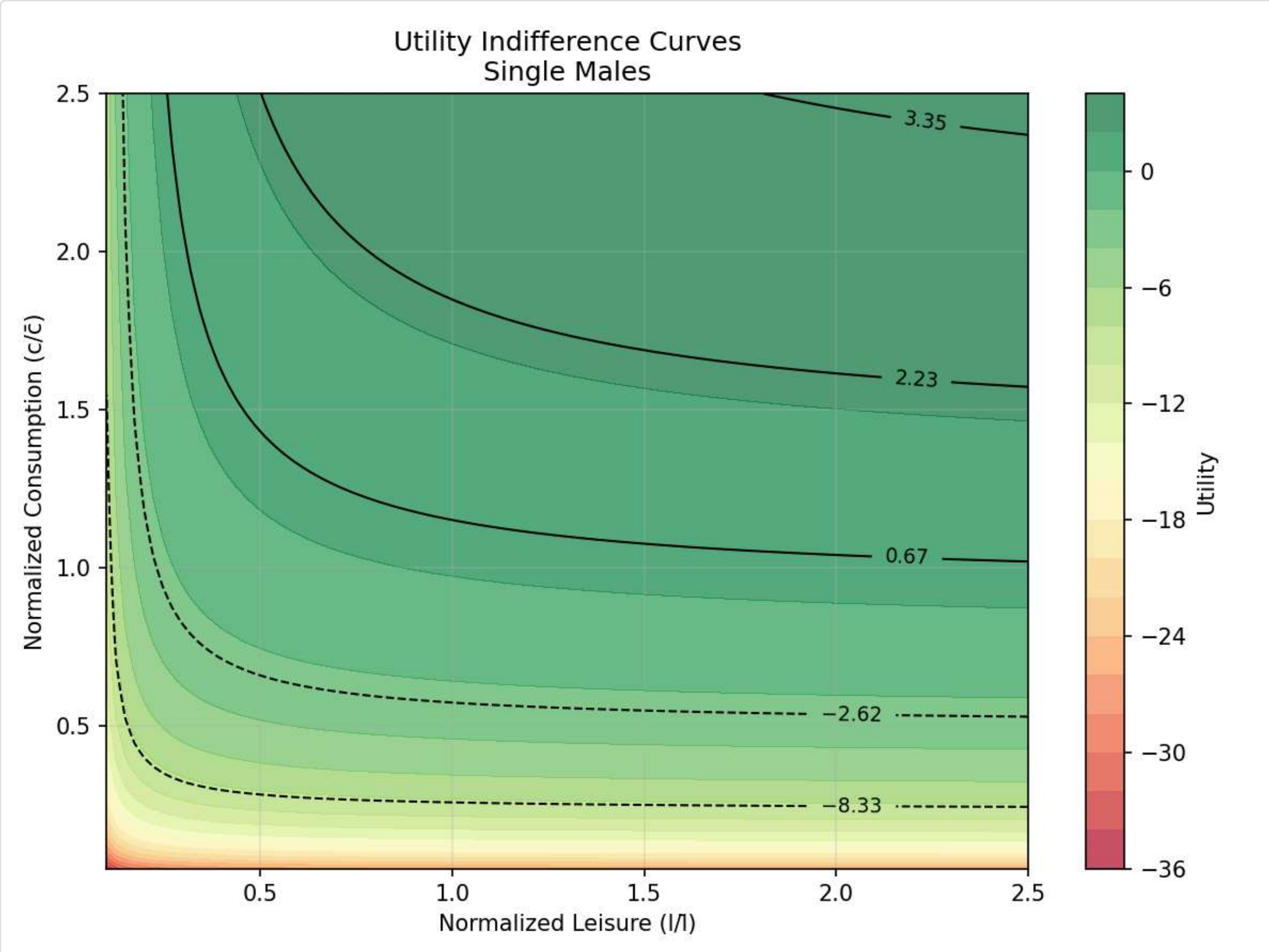


Females in Couples - MUC & MUL Distributions

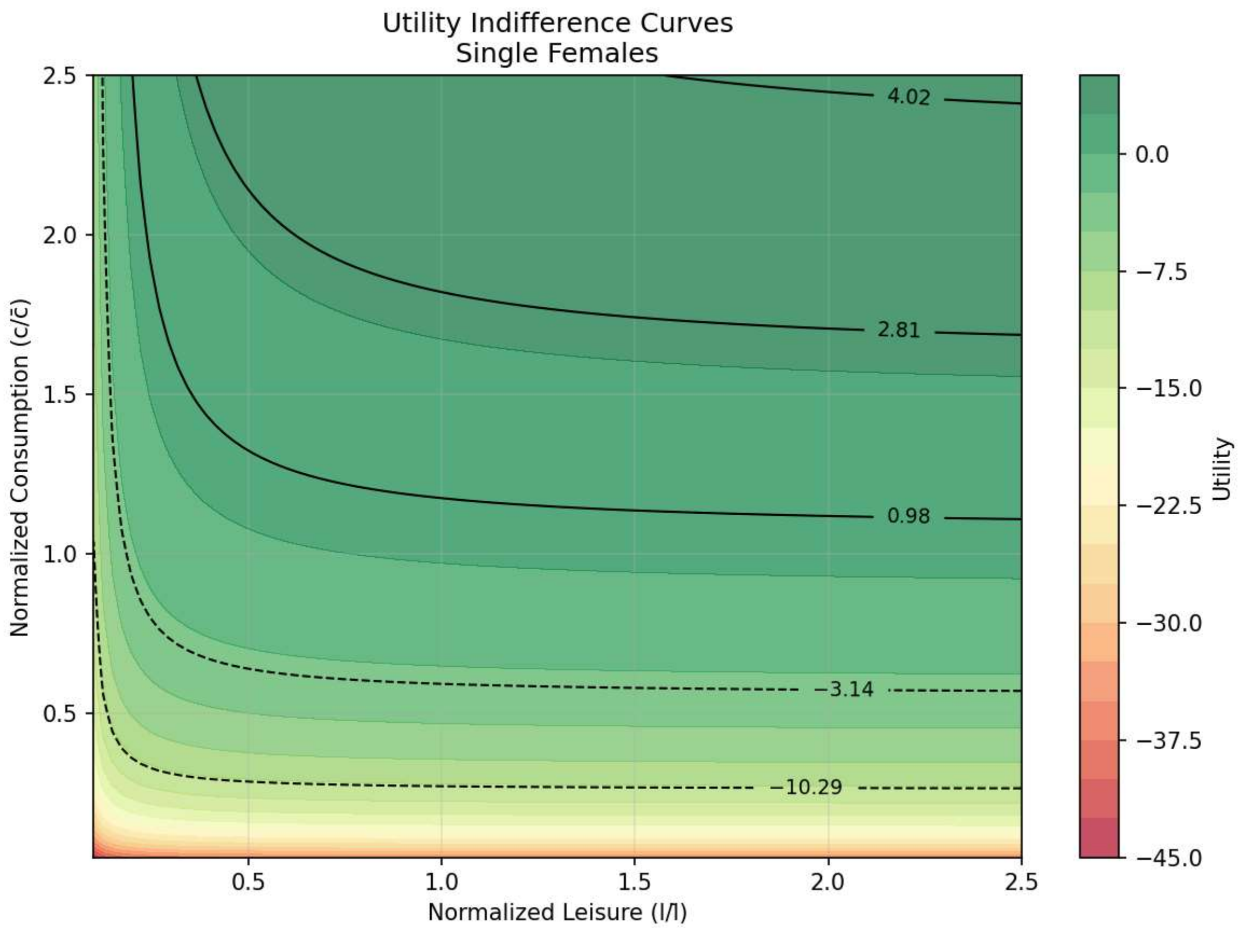


Utility Indifference Curves by Group

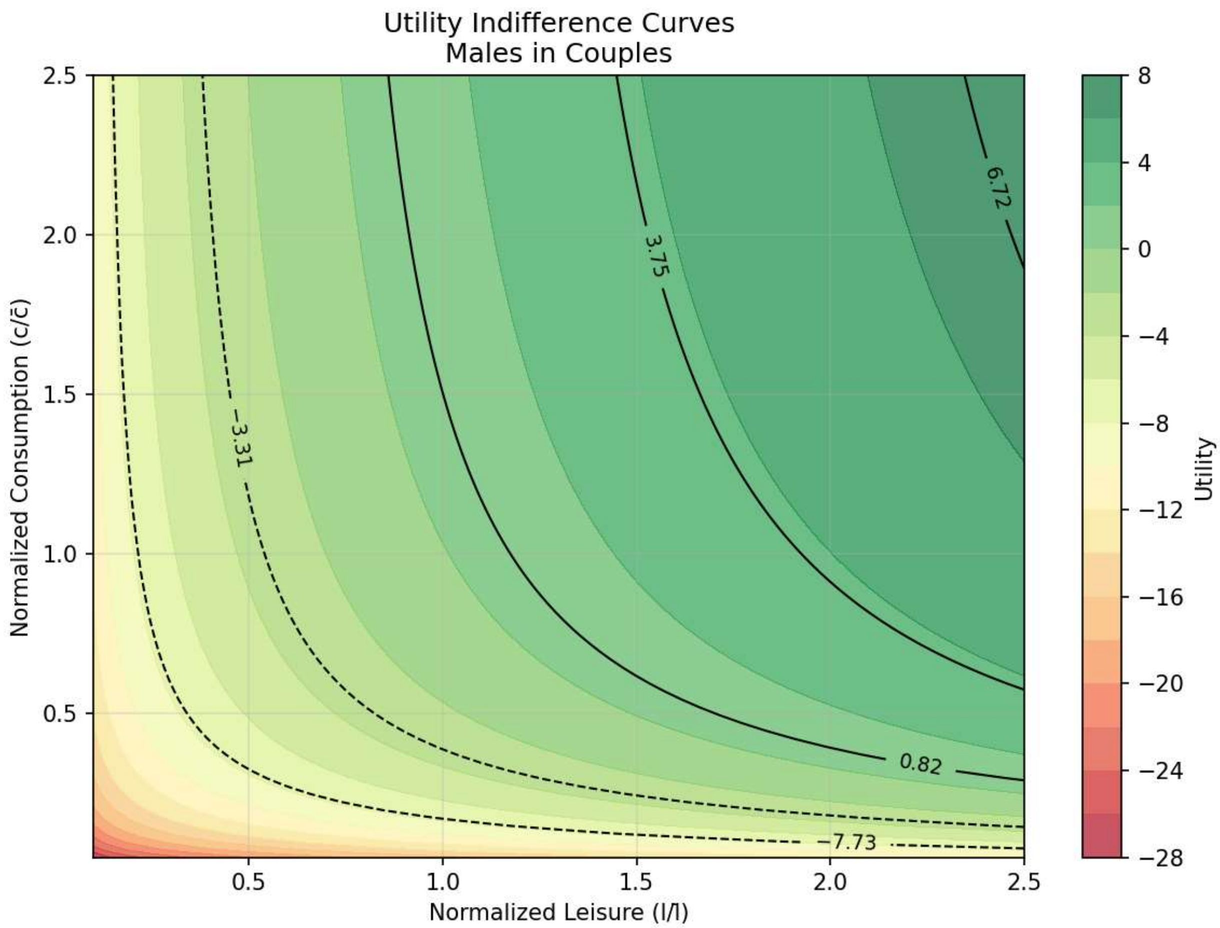
Single Males



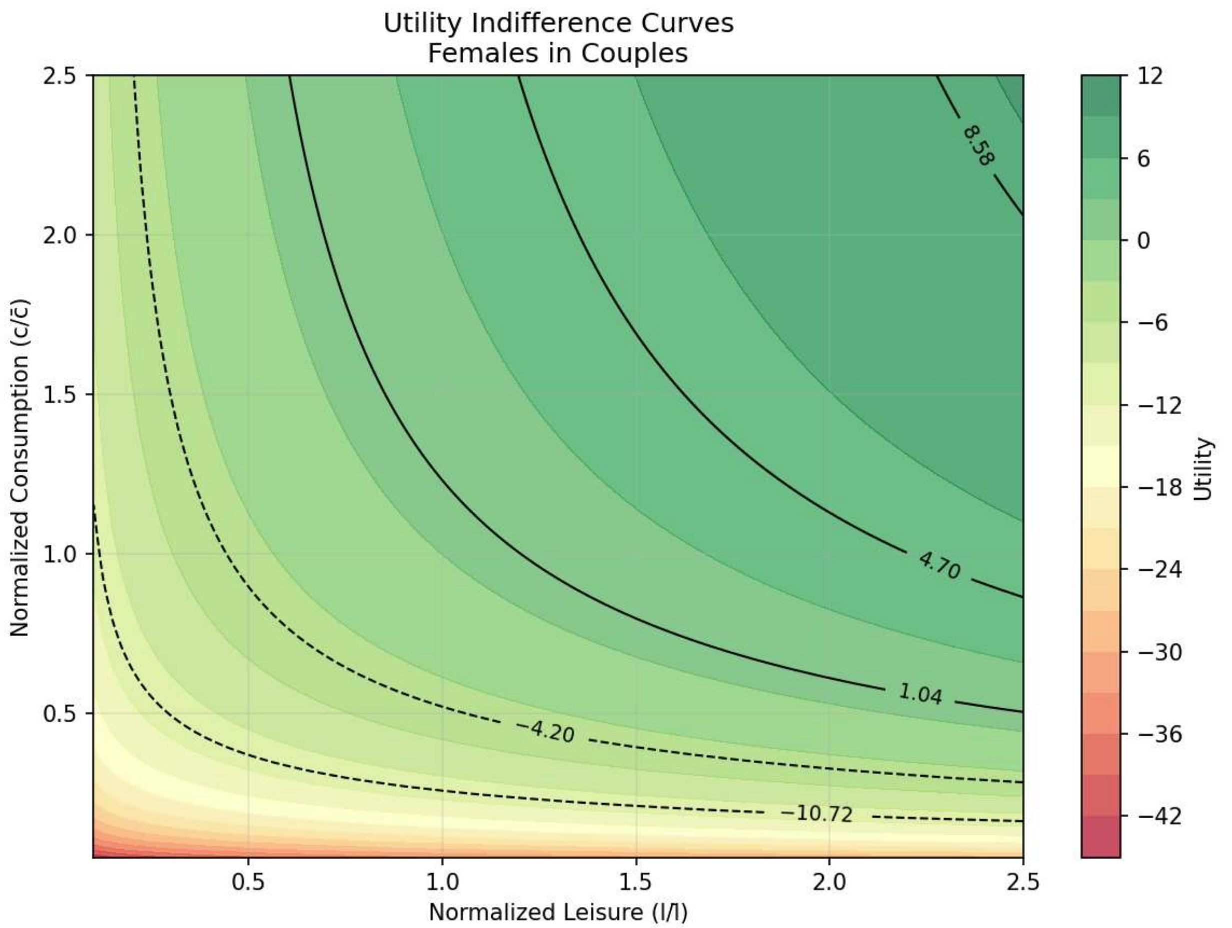
Single Females



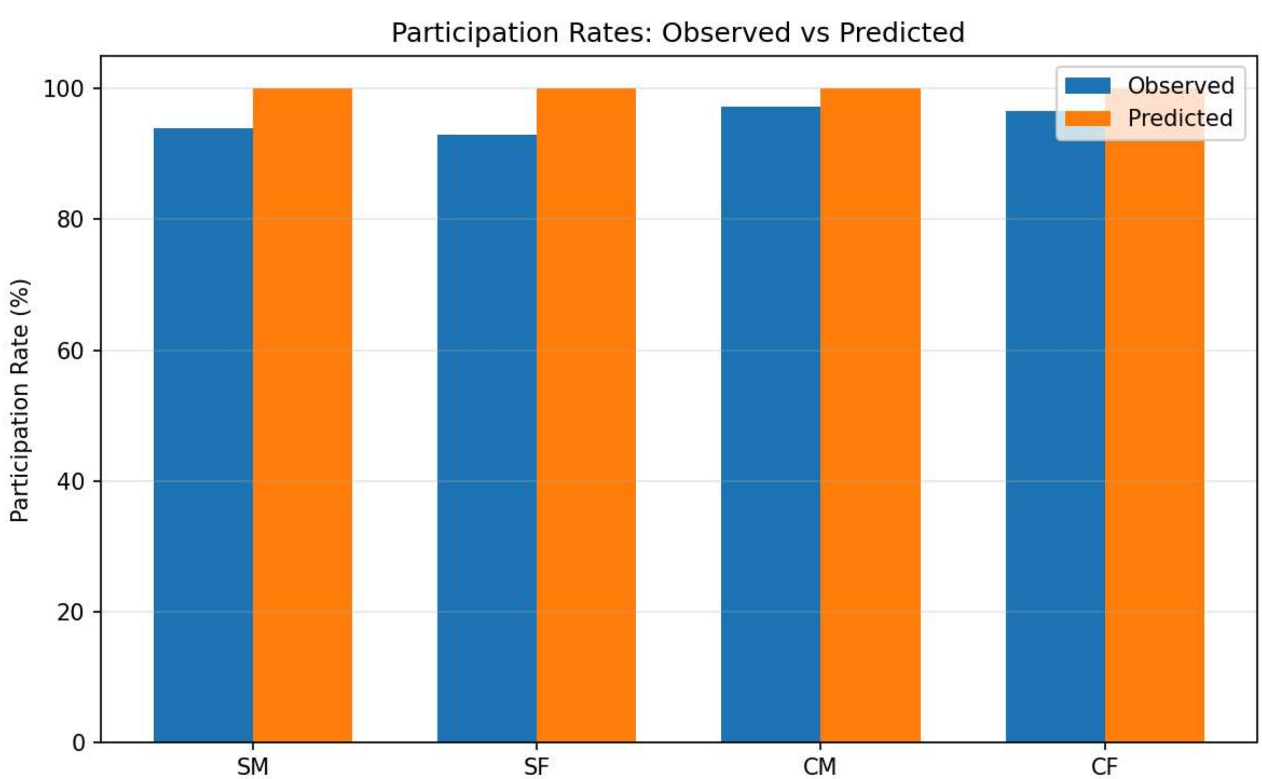
Males in Couples



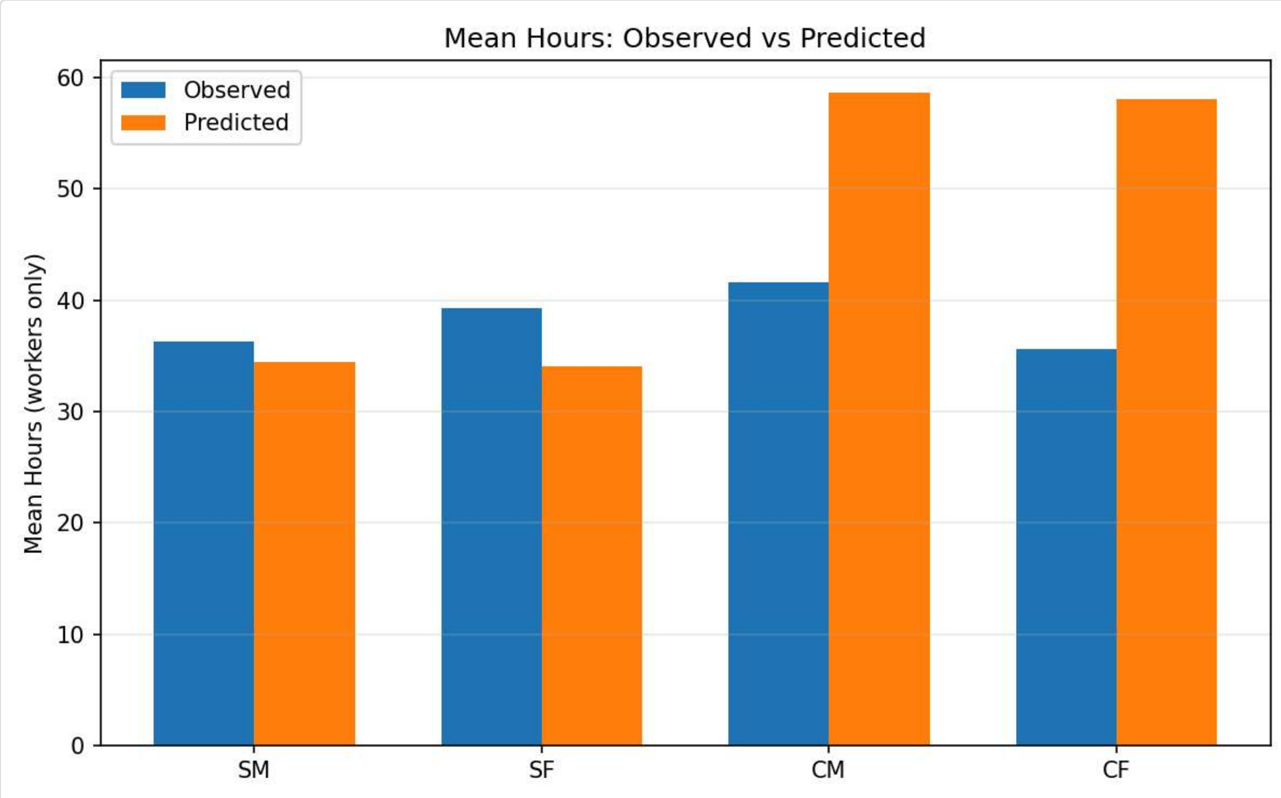
Females in Couples



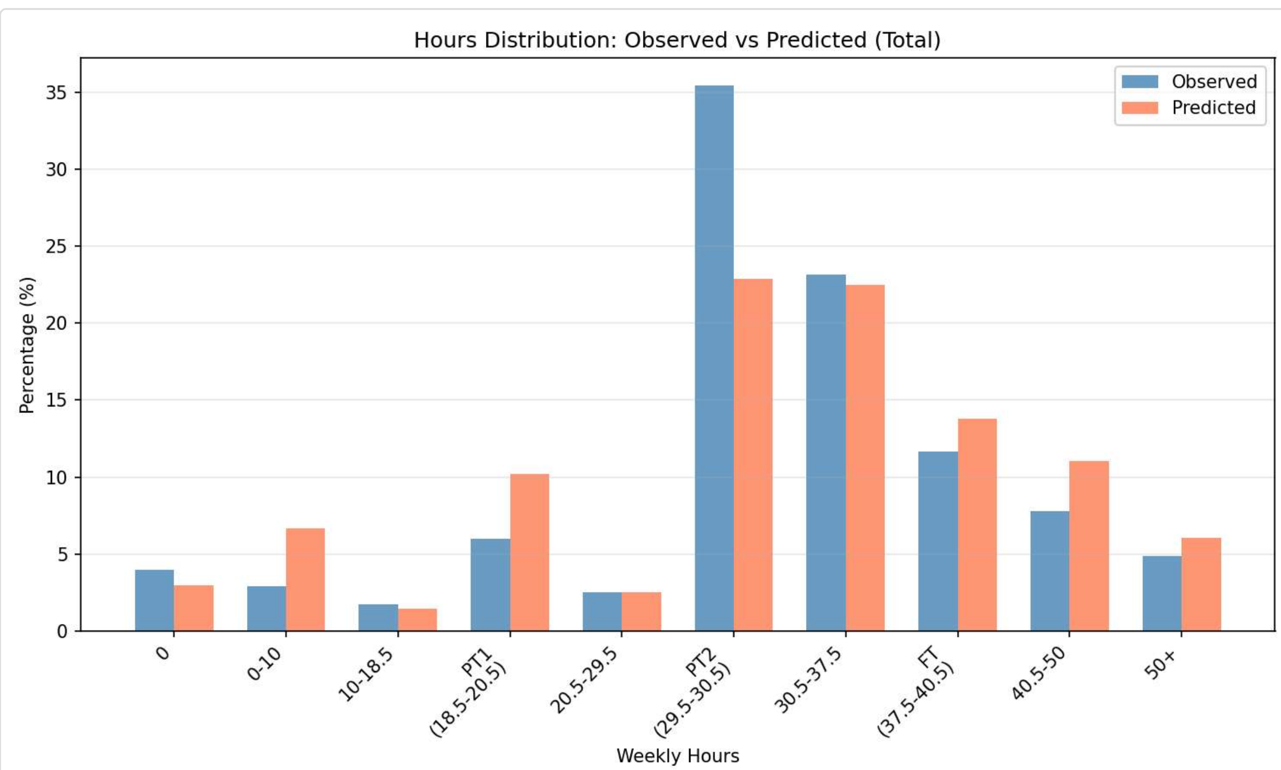
Participation

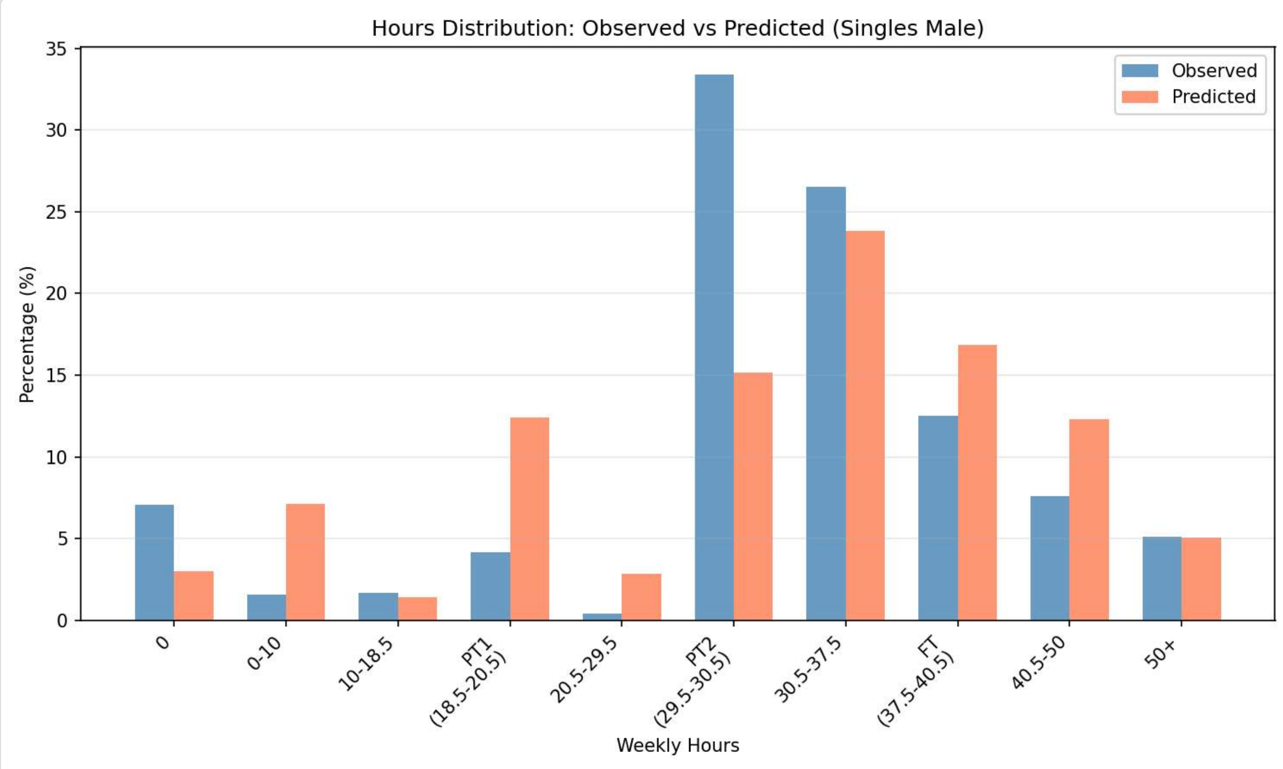


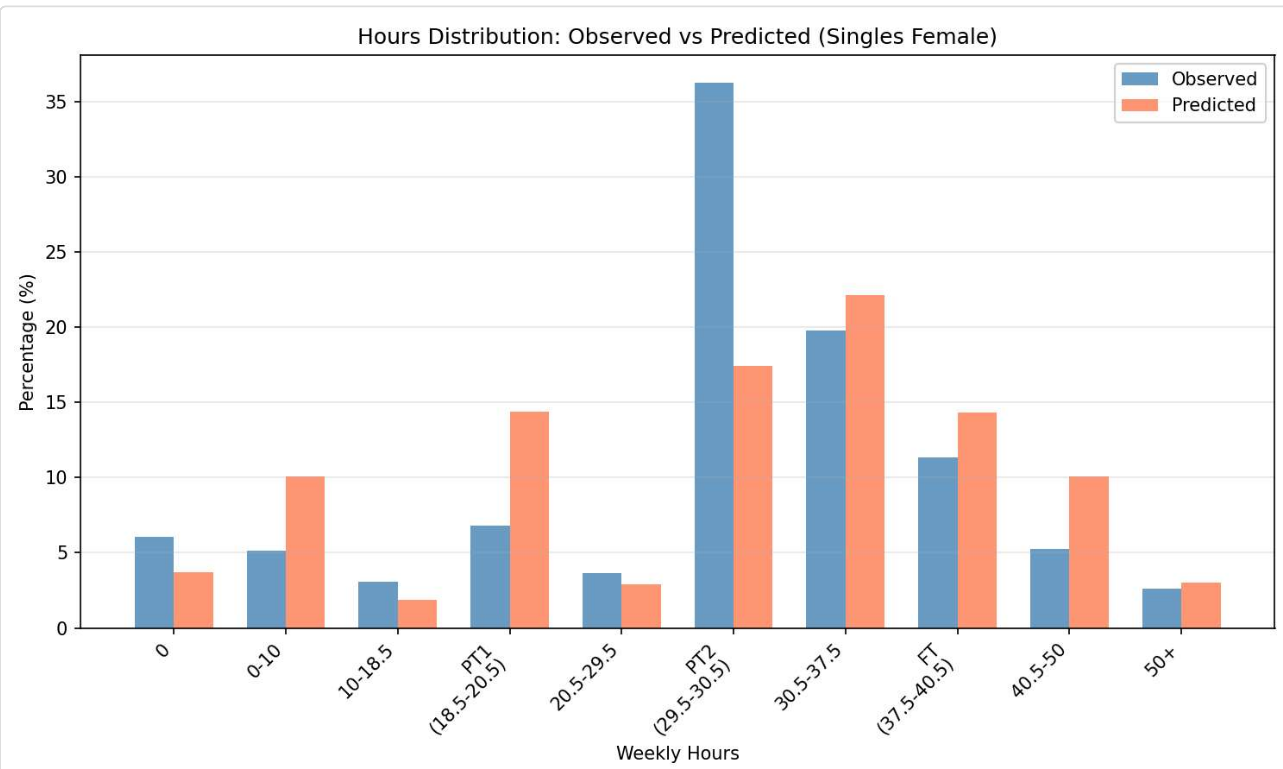
Mean Hours

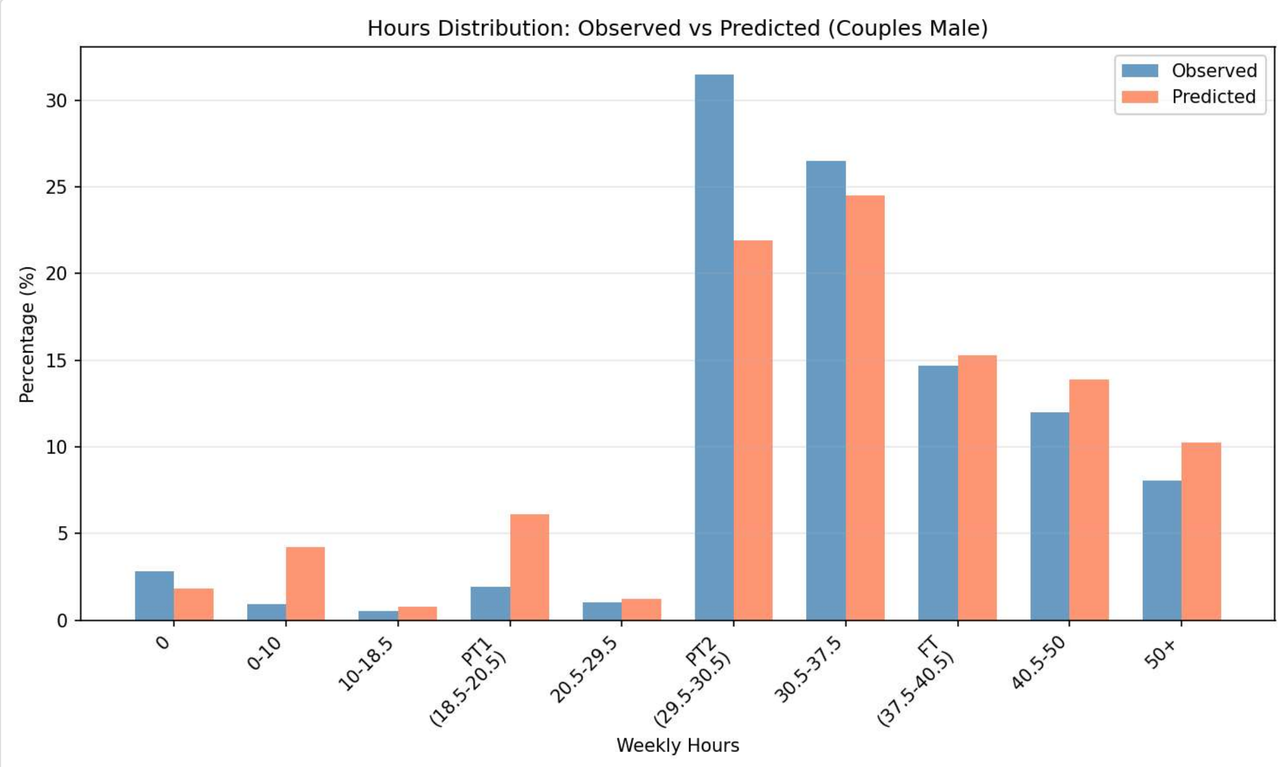


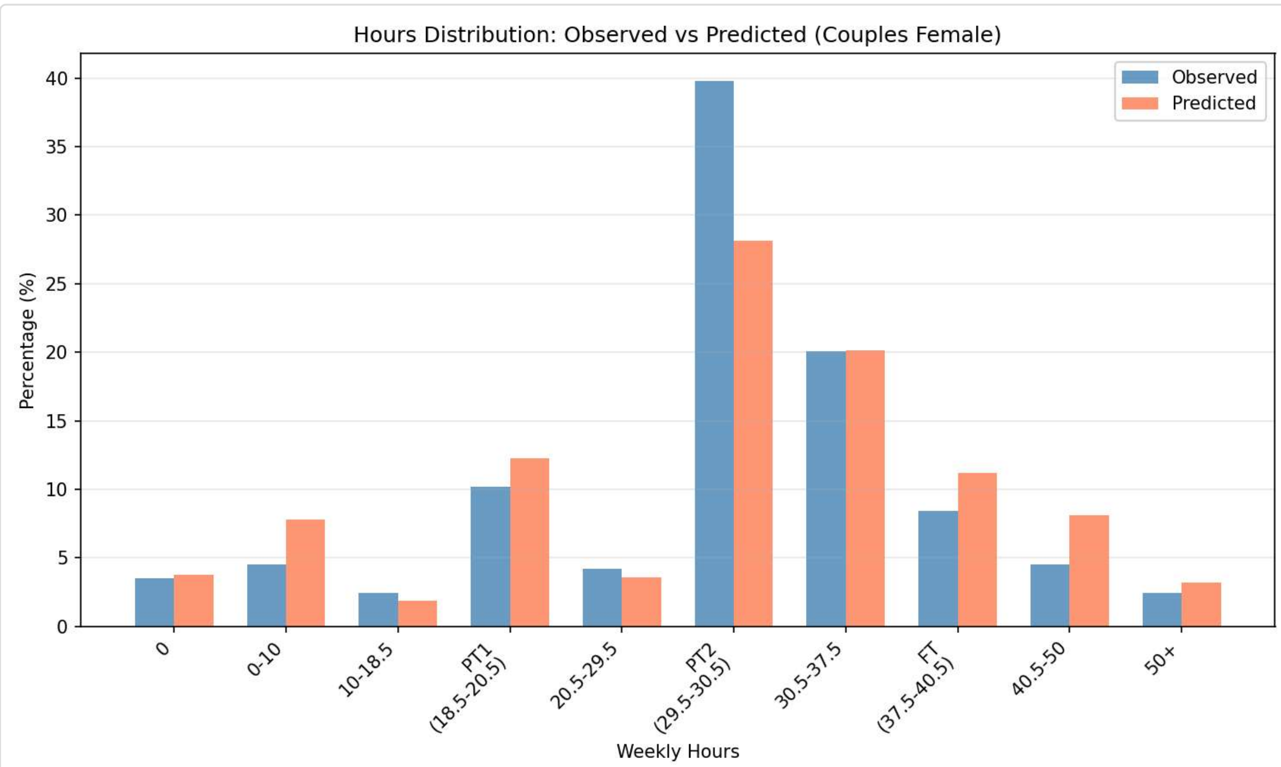
?? Hours Distribution: Observed vs Predicted



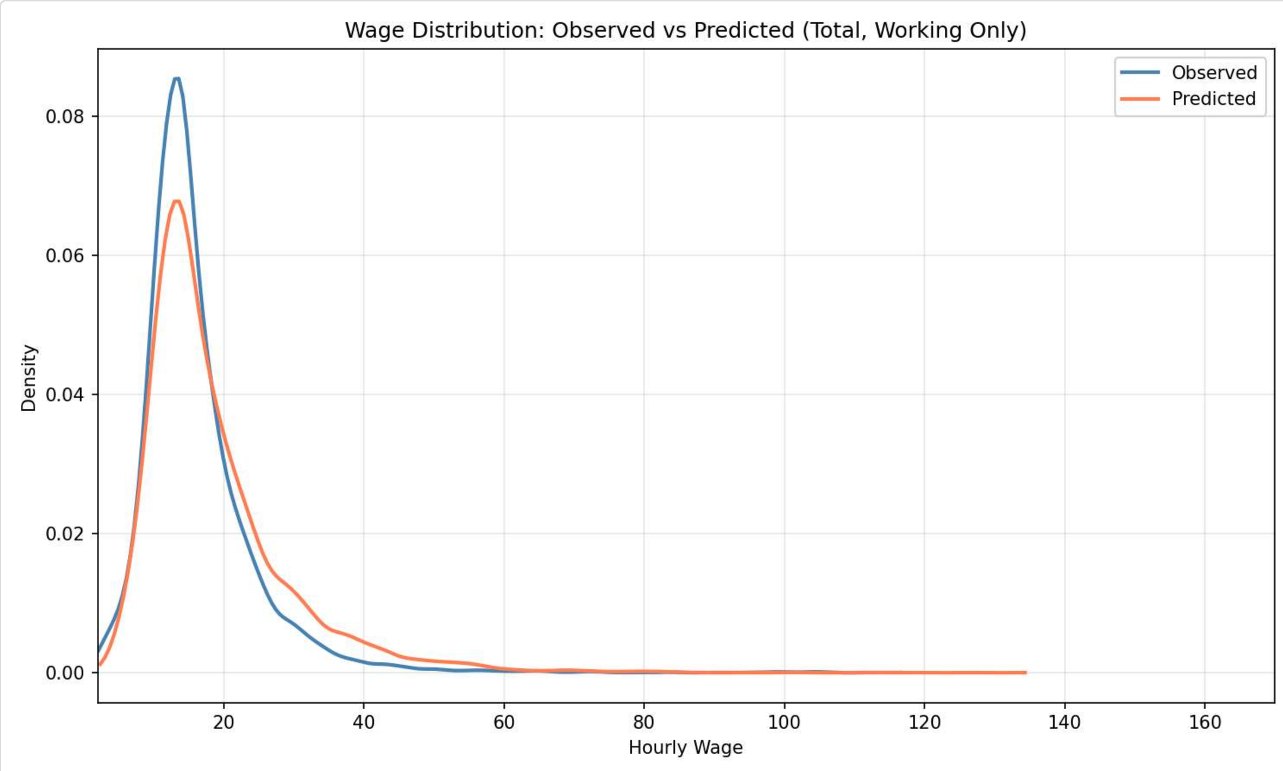

Hours Distribution: Observed vs Predicted (Singles Male)
Observed
Predicted
Percentage (%)
Weekly Hours
0
0-10
10-18.5
PT1 (18.5-20.5)
20.5-29.5
PT2 (29.5-30.5)
30.5-37.5
FT (37.5-40.5)
40.5-50
50+

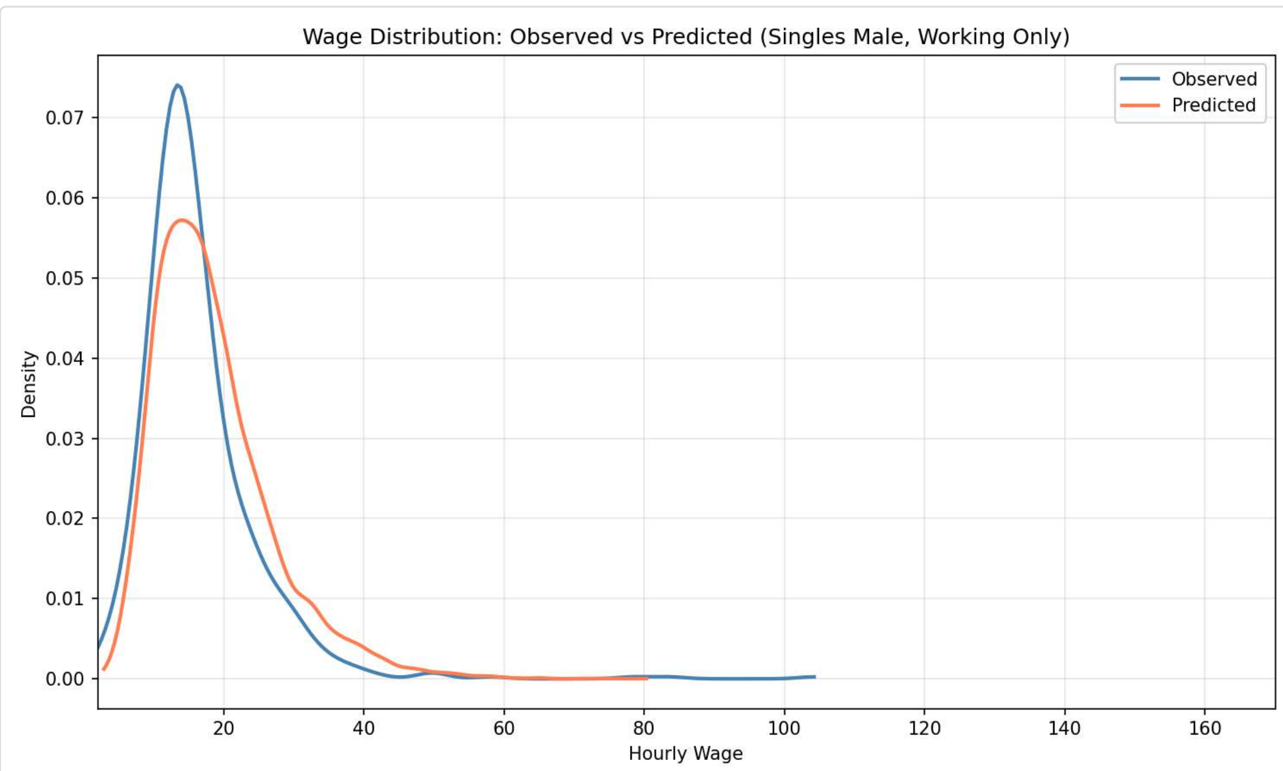

Hours Distribution: Observed vs Predicted (Singles Female)
Observed
Predicted
Percentage (%)
Weekly Hours
0
0-10
10-18.5
PT1 (18.5-20.5)
20.5-29.5
PT2 (29.5-30.5)
30.5-37.5
FT (37.5-40.5)
40.5-50
50+

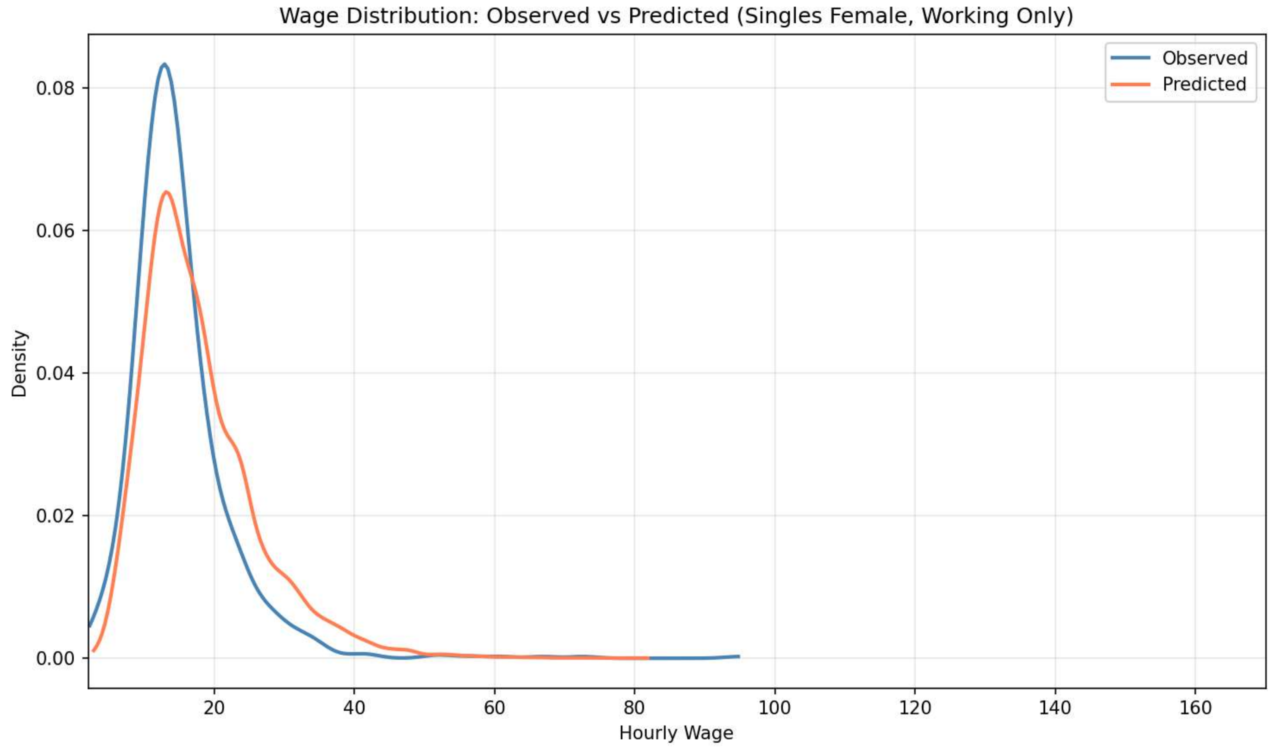




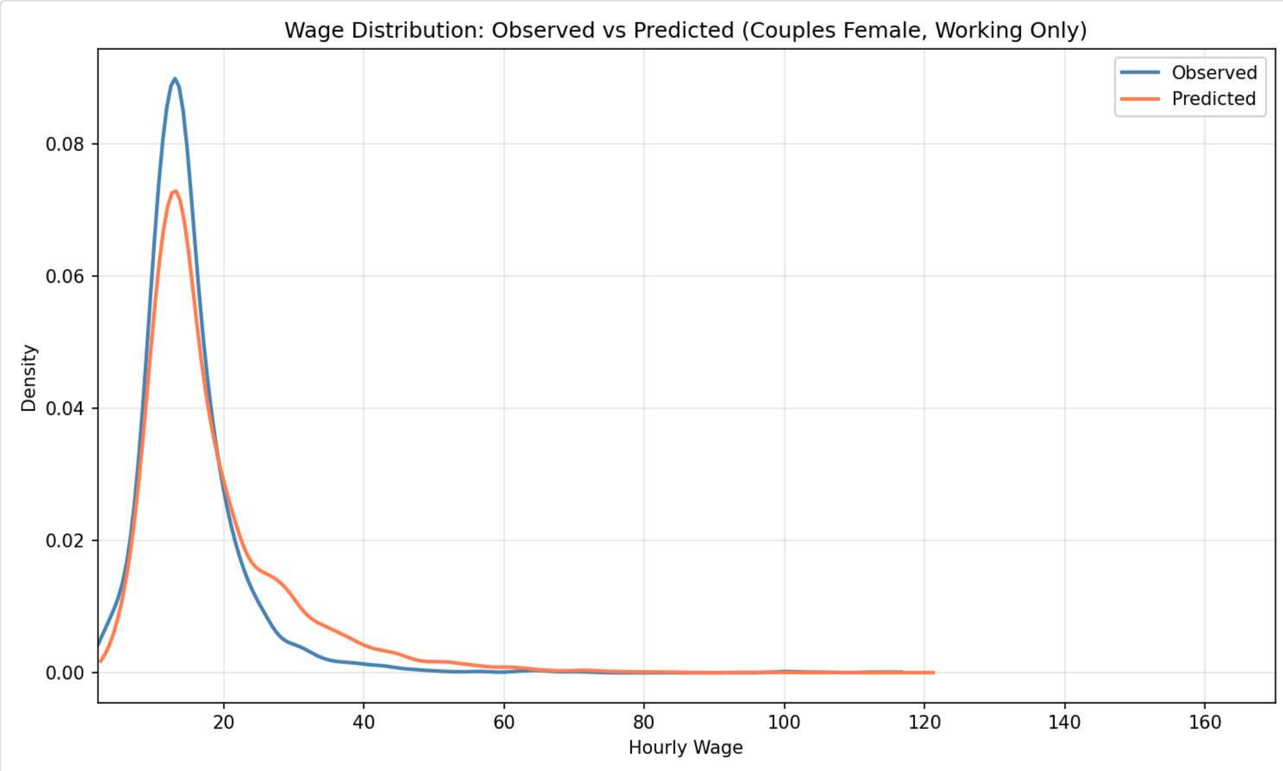
?? Wage Distribution: Observed vs Predicted (Working Only)


Wage Distribution: Observed vs Predicted (Total, Working Only)
Observed
Predicted
Density
Hourly Wage
0.00
0.02
0.04
0.06
0.08
20
40
60
80
100
120
140
160


Wage Distribution: Observed vs Predicted (Singles Male, Working Only)
Observed
Predicted
Density
Hourly Wage
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
20
40
60
80
100
120
140
160


Wage Distribution: Observed vs Predicted (Singles Female, Working Only)
Observed
Predicted
Density
0.08
0.06
0.04
0.02
0.00
20
40
60
80
100
120
140
160
Hourly Wage


Wage Distribution: Observed vs Predicted (Couples Male, Working Only)
Observed
Predicted
Density
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
20
40
60
80
100
120
140
160
Hourly Wage



Group-Specific Parameters

Single Males

beta_E	-2.6148
beta_E_educH	0.2575
beta_E_gsur	-0.7719
beta_c	0.7175
beta_h_ft	1.4632
beta_h_pt1	-0.5203
beta_h_pt2	0.3721
beta_l0	4.4666
beta_l_age	0.0008
beta_l_age2	0.0018
beta_l_educH	-1.3035

Single Females

beta_E	-2.6148
beta_E_educH	0.2575
beta_E_gsur	-0.7719
beta_c	0.4630
beta_h_ft	1.4632
beta_h_pt1	-0.5203
beta_h_pt2	0.3721
beta_l0	5.7585
beta_l_age	-0.0165
beta_l_age2	0.0033
beta_l_educH	-2.2710

beta_occ_2	-1.5129
beta_occ_2_cf	0.2050
beta_occ_2_cm	-1.4707
beta_occ_3	-2.1726
beta_occ_3_cf	-0.1904
beta_occ_3_cm	-2.2061
beta_occ_4	0.0175
beta_occ_4_cf	1.1460
beta_occ_4_cm	0.4862
beta_w0	2.0411
beta_w_educH	0.3055
beta_w_educL	-0.0476
beta_w_pexp	0.0174
beta_w_pexp2	-0.0002
sigma	0.4232
theta_c	-0.8560
theta_l	-0.7038

beta_l_nkids	-0.0496
beta_occ_2	-0.0147
beta_occ_2_cf	0.2050
beta_occ_2_cm	-1.4707
beta_occ_3	-0.5745
beta_occ_3_cf	-0.1904
beta_occ_3_cm	-2.2061
beta_occ_4	0.7934
beta_occ_4_cf	1.1460
beta_occ_4_cm	0.4862
beta_w0	2.0411
beta_w_educH	0.3055
beta_w_educL	-0.0476
beta_w_pexp	0.0174
beta_w_pexp2	-0.0002
sigma	0.4232
theta_c	-1.0884
theta_l	-0.7238

Males in Couples

beta_E	-2.6148
beta_E_educH	0.2575
beta_E_gsur	-0.7719
beta_c	5.2620
beta_h_ft	1.4632
beta_h_pt1	-0.5203
beta_h_pt2	0.3721
beta_l0	2.7023
beta_l_age	-0.0074
beta_l_age2	0.0013
beta_l_educH	-0.9259

Females in Couples

beta_E	-2.6148
beta_E_educH	0.2575
beta_E_gsur	-0.7719
beta_c	5.2620
beta_h_ft	1.4632
beta_h_pt1	-0.5203
beta_h_pt2	0.3721
beta_l0	5.6972
beta_l_age	-0.0578
beta_l_age2	0.0016
beta_l_educH	-1.4890

beta_occ_2_cf	0.2050
beta_occ_2_cm	-1.4707
beta_occ_3_cf	-0.1904
beta_occ_3_cm	-2.2061
beta_occ_4_cf	1.1460
beta_occ_4_cm	0.4862
beta_w0	2.0411
beta_w_educH	0.3055
beta_w_educL	-0.0476
beta_w_pexp	0.0174
beta_w_pexp2	-0.0002
sigma	0.4232
theta_c	0.2153
theta_l	-0.7330

beta_l_nkids	0.2149
beta_occ_2_cf	0.2050
beta_occ_2_cm	-1.4707
beta_occ_3_cf	-0.1904
beta_occ_3_cm	-2.2061
beta_occ_4_cf	1.1460
beta_occ_4_cm	0.4862
beta_w0	2.0411
beta_w_educH	0.3055
beta_w_educL	-0.0476
beta_w_pexp	0.0174
beta_w_pexp2	-0.0002
sigma	0.4232
theta_c	0.2153
theta_l	-0.6957

📋 Parameter Estimates by Category

*Significance: *** p<0.001, ** p<0.01, * p<0.05*

☐ Bounded (has constraints) ☐ ⚠ Hit bound ☐ Warning: degenerate SE (near zero) ☐ p ∈ [0.05, 0.1)
☐ p ∈ [0.1, 0.25) ☐ p ≥ 0.25

🎯 Preference Parameters (Utility Function)

Parameters determining the utility from consumption (β_c, θ_c) and leisure (β_l, θ_l) by demographic group.*

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.beta_l0_sm	4.4666	0.1590	28.09	0.0000 ***	0.0500	50.0000	1.0000
joint.beta_l_age_sm	0.0008	0.0246	0.03	0.9726	-5.0000	5.0000	0.0000

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.beta_l_age2_sm	0.0018	0.0020	0.87	0.3828	-1.0000	1.0000	0.0000
joint.beta_l_educH_sm	-1.3035	0.3724	-3.50	0.0005 ***	-8.0000	5.0000	0.0000
joint.beta_c_sm	0.7175	0.2660	2.70	0.0070 **	0.0500	50.0000	1.0000
joint.theta_l_sm	-0.7038	0.0990	-7.11	0.0000 ***	-8.0000	0.9500	-1.0000

⏰ Employment and Hours Opportunity Parameters

Working-gated employment intercept and hours focal-point indicators, plus residual market-access shifters (gsur, education) declared in `market_opportunity.shifters`.

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.beta_E	-2.6148	0.3022	-8.65	0.0000 ***	-25.0000	25.0000	0.0000
joint.beta_h_pt1	-0.5203	0.1086	-4.79	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_h_pt2	0.3721	0.1110	3.35	0.0008 ***	-10.0000	10.0000	0.0000
joint.beta_h_ft	1.4632	0.0499	29.31	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_E_gsur	-0.7719	0.2213	-3.49	0.0005 ***	-10.0000	10.0000	0.0000
joint.beta_E_educH	0.2575	0.2409	1.07	0.2852	-10.0000	10.0000	0.0000

💰 Wage Opportunity / Mincer Parameters

Log-normal wage opportunity: $\mu_W(X)$ intercept, education, experience, and residual standard deviation σ.

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.beta_w0	2.0411	0.0265	77.00	0.0000 ***	-10.0000	20.0000	2.0000
joint.beta_w_educL	-0.0476	0.0209	-2.27	0.0230 *	-5.0000	5.0000	-0.1000
joint.beta_w_educH	0.3055	0.0150	20.36	0.0000 ***	-5.0000	5.0000	0.2000
joint.beta_w_pexp	0.0174	0.0022	7.83	0.0000 ***	-1.0000	1.0000	0.0200
joint.beta_w_pexp2	-0.0002	0.0000	-4.19	0.0000 ***	-0.1000	0.1000	-0.0003
joint.sigma	0.4232	0.0041	103.43	0.0000 ***	0.1000	20.0000	0.5000

🧭 Occupation Opportunity Parameters

Working-gated occupation shifters declared in `occupation_opportunity.shifters`; routed by `applies_to` group (sm/sf/cm/cf). Reference category is the YAML's `occupation_opportunity.reference`.

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.beta_occ_2_sm	-1.5129	0.1423	-10.63	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_3_sm	-2.1726	0.1845	-11.77	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_4_sm	0.0175	0.0861	0.20	0.8387	-10.0000	10.0000	0.0000
joint.beta_occ_2_sf	-0.0147	0.1129	-0.13	0.8961	-10.0000	10.0000	0.0000
joint.beta_occ_3_sf	-0.5745	0.1290	-4.45	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_4_sf	0.7934	0.0922	8.60	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_2_cm	-1.4707	0.1139	-12.91	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_3_cm	-2.2061	0.1478	-14.93	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_4_cm	0.4862	0.0687	7.08	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_2_cf	0.2050	0.1007	2.04	0.0418 *	-10.0000	10.0000	0.0000
joint.beta_occ_3_cf	-0.1904	0.1115	-1.71	0.0878	-10.0000	10.0000	0.0000

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